Ordinary Differential Equations

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1 Linear and Non-Linear Equations

For a linear initial-value problem (IVP),

$$\frac{dy}{dt} = -p(t)y + g(t), \qquad y(t_0) = y_0.$$

More generally, an IVP may have the form

$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0.$$

Two questions arise:

1. Does the IVP have at least one solution (existence)?
2. If so, is that solution unique?

1.1 Existence and Uniqueness for Linear Equations

If p and g are continuous on an interval $I = (\alpha, \beta)$ containing t_0, then the linear IVP has a unique solution defined throughout I.

With

$$\mu(t) = \exp\left(\int_{t_0}^{t} p(s)\, ds\right),$$

the solution can be written as

$$y(t) = \frac{y_0 + \displaystyle\int_{t_0}^{t} \mu(s) g(s)\, ds}{\mu(t)}.$$

For uniqueness, suppose u and v satisfy the same equation and initial condition. Their difference $w = u - v$ satisfies

$$w' = -p(t)w, \qquad w(t_0) = 0.$$

Multiplying by $\mu(t)$ gives

$$\frac{d}{dt}\big(\mu(t)w(t)\big) = 0.$$

Consequently, $\mu(t)w(t) = \mu(t_0)w(t_0) = 0$. Since $\mu(t) > 0$ throughout I, we have $w = 0$, so $u = v$. The formula above also proves existence: differentiating it recovers the differential equation and setting $t = t_0$ gives $y(t_0) = y_0$.

1.2 Existence and Uniqueness for Nonlinear Equations

For $y' = f(t, y)$, continuity of f near the initial point guarantees a solution at least locally. If f and $\partial f/\partial y$ are continuous on a rectangle around that point, the local solution is unique. These are local statements: a solution may cease to exist at a finite endpoint of its maximal interval.

1.3 Example: A Linear IVP

Does $ty' + 2y = 4t^2$, $y(1) = 2$, have a unique solution?

Solution.

For $t \neq 0$, divide by t:

$$y' + \frac{2}{t}y = 4t.$$

On $I = (0, \infty)$, both coefficients are continuous. The integrating factor is $\mu(t) = t^2$, so

$$(t^2 y)' = 4t^3, \qquad t^2 y = t^4 + C.$$

Thus $y = t^2 + C/t^2$. The initial condition gives $C = 1$, hence

$$\boxed{y(t) = t^2 + \frac{1}{t^2}, \qquad 0 < t < \infty.}$$

The linear theorem guarantees uniqueness on $(0, \infty)$. This solution cannot be extended through $t = 0$, where it diverges.

1.4 Example: A Nonlinear IVP with a Singular Boundary

Consider

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y - 1)}, \qquad y(0) = -1.$$

Solution.

The right-hand side and its derivative with respect to y are continuous near $(0, -1)$, since $y \neq 1$ there. A unique local solution therefore exists. Separating and integrating yields

$$2(y - 1)\, dy = (3x^2 + 4x + 2)\, dx, \qquad (y - 1)^2 = x^3 + 2x^2 + 2x + C.$$

Since $y(0) = -1$, we have $C = 4$. Taking the negative square-root branch to match the initial value gives

$$\boxed{y(x) = 1 - \sqrt{(x + 2)(x^2 + 2)}, \qquad -2 < x < \infty.}$$

At $x = -2$, this formula reaches $y = 1$, where the original equation is undefined. Thus $(-2, \infty)$ is the maximal interval containing 0 on which this solution satisfies the IVP.

1.5 Example: Existence Without Uniqueness

Consider

$$\frac{dy}{dt} = y^{1/3}, \qquad y(0) = 0, \quad t \geq 0.$$

Solution.

The function $f(y) = y^{1/3}$ is continuous everywhere, so a local solution exists. However, $f'(y) = \frac{1}{3}y^{-2/3}$ is undefined at $y = 0$, so the usual uniqueness criterion does not apply. Indeed, both

$$y(t) = 0 \qquad \text{and} \qquad y(t) = \left(\frac{2t}{3}\right)^{3/2}$$

satisfy the IVP for $t \geq 0$. More generally, for any $a \geq 0$,

$$y_a(t) = \begin{cases} 0, & 0 \leq t \leq a, \\ \left(\dfrac{2(t-a)}{3}\right)^{3/2}, & t > a \end{cases}$$

is a solution. The derivative from the right at $t = a$ is zero, so these pieces join differentiably. There are infinitely many solutions.

1.6 Example: Unique Locally, but Not for All Time

Consider

$$\frac{dy}{dx} = y^2, \qquad y(0) = 1.$$

Solution.

Both $f(x, y) = y^2$ and $\partial f/\partial y = 2y$ are continuous everywhere, so the IVP has a unique local solution. Separating gives

$$-\frac{1}{y} = x + C.$$

Applying $y(0) = 1$ gives

$$\boxed{y(x) = \frac{1}{1-x}, \qquad -\infty < x < 1.}$$

The solution grows without bound as $x \to 1^-$. Its maximal interval containing the initial point is $(-\infty, 1)$, even though the equation's right-hand side is defined for every (x, y).